

Ephys recordings with AstroDREADD/DPCPX dataset “SeqBias_AstroBlock_NPX” Overview

Electrophysiology recordings from PFC or MC neuronal populations during a sensory graded go/no-go task. Data include sessions with chemogenetic astrocytes blockade (Gq-DREADD in astrocytes + CNO injection), blockade of astrocyte-neuron purinergic signaling (DPCPX topical application), and saline injection controls.

Unique features

- High-throughput electrophysiological recordings from **PFC or MC/Striatum** during **complex behavior**
- Electrophysiological recordings with **chemogenetic blockade of astrocytes** dynamics or **astrocyte-neuronal signaling** preventing behavioral optimization after surprising outcomes

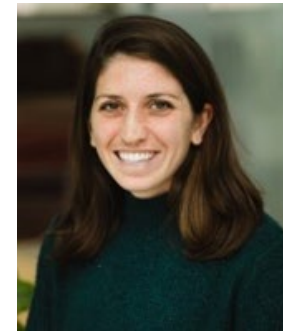
Caveats

- Variable number of sessions per animal
- Same control group for Astro-DREADD and DPCPX manipulations
- Most recording sessions from PFC

Publications

- *Drummond G., et al. Cortical norepinephrine-astrocyte signaling critically mediates learned behavior. biorXiv (2024)* [see also the “SeqBias_AstroNE_imaging” dataset]

Experimenters:

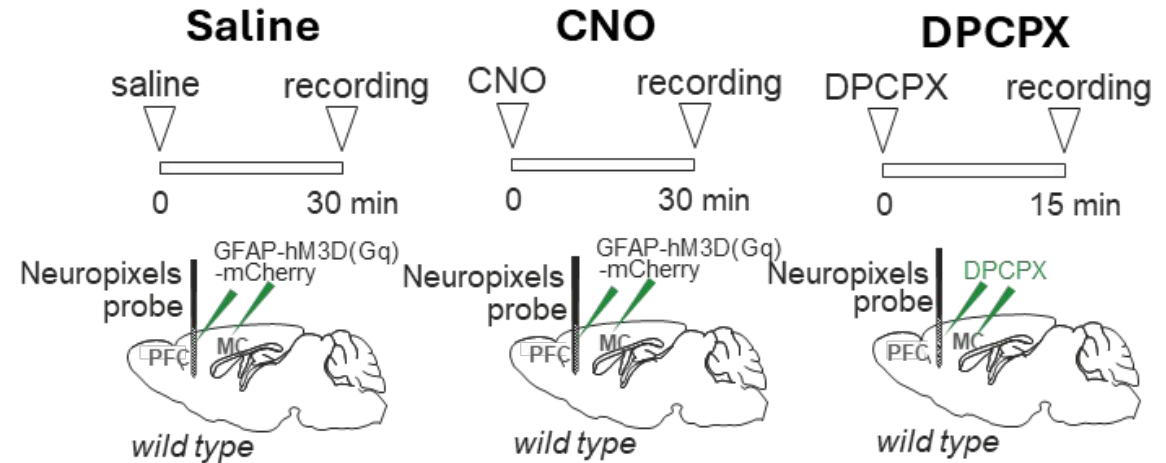
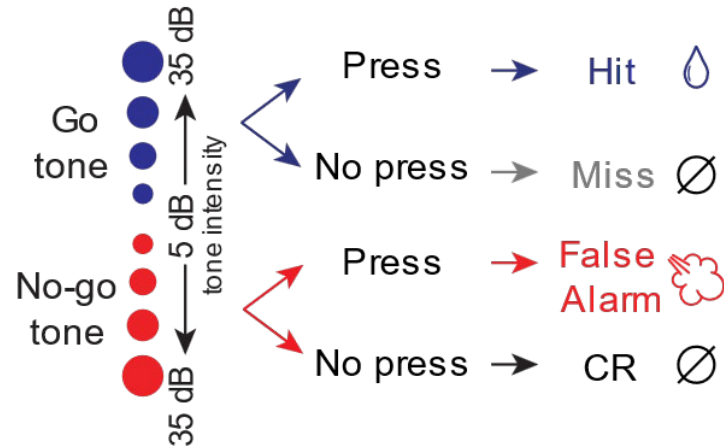


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Experiment design



Behavioral task: sensory graded go/no-go task. On each trial mice were presented an auditory stimulus, either a go tone (12 kHz) or a no-go tone (4 kHz). After the onset of the 0.5s sound stimulus, mice had 0.8s to respond or hold still. If they pressed the lever during a go trial, they received a water reward. If they pressed during a no-go trial, they received a mild air puff punishment. Absence of response on go trials (miss) or not pressing the lever during no-go trials (correct rejections) were not reinforced. To vary the level of stimulus evidence, 4 intensities were used per frequency for a total of 8 different stimuli. On trials after a false alarm, mice showed increased behavioral performance. See *Breton-Provencher et al., Nature (2022)* for full details.

Chemogenetic manipulations: neuronal activity recorded from the same mice with either saline, CNO activation of Gq-DREADD in cortical astrocytes, or topical application of the A1 receptor antagonist DPCPX; using the same mice with enough wash-out between experiments enabling direct comparison between manipulations.

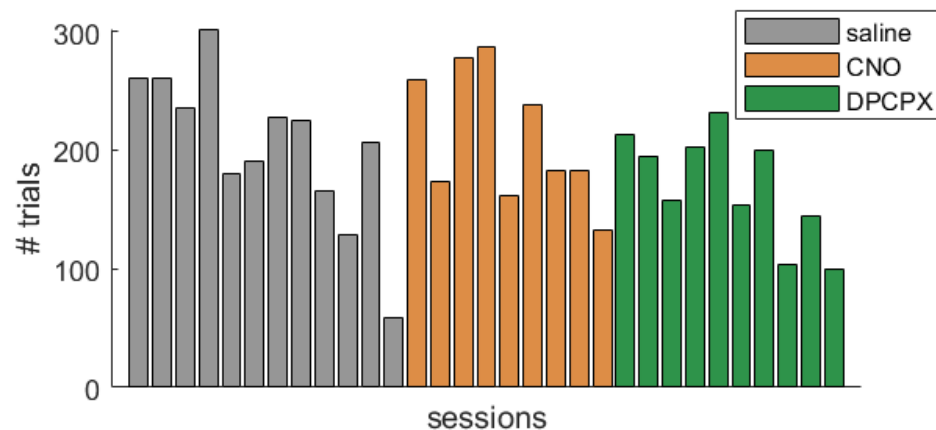
Ephys recordings with AstroDREADD/DPCPX dataset “SeqBias_AstroBlock_NPX” Recordings and dataset Ns

High-throughput **Neuropixel 1.0** (384 channels)
recordings from PFC or MC/Striatum

Included **data**: spikeTimes, spikeWaves, lfp

Total of **M** = 1647 **neurons** recorded across **n** = 31
sessions from **N** = 8 **mice**.

Trials per session (discarding early presses)



exp. condition

	recording area		
	PFC	MC	Striatum
Saline	M = 574 n = 8 N = 6	M = 97 n = 4 N = 4	M = 56 n = 4 N = 4
CNO	M = 360 n = 6 N = 5	M = 61 n = 3 N = 3	M = 43 n = 3 N = 3
DPCPX	M = 429 n = 9 N = 5	M = 16 n = 1 N = 1	M = 11 n = 1 N = 1

Number of cells, sessions and mice per
area and experimental condition

Ephys recordings with AstroDREADD/DPCPX dataset “SeqBias_AstroBlock_NPX” Metadata

The dataset misses information about mice age at the time of each recording (set to unknown, “U”). All other required metadata are present.

Specific metadata keywords for this dataset:

sessionInfo:

condition: unique identifiers for experimental condition are “saline” (Astrocytes DREADD + saline injection), “CNO” (Astrocytes DREADD + CNO injection) or “DPCPX” (topical DPCPX application)

Data preprocessing:

Spike sorting: Raw neuropixel data were bandpassed filtered 300-9000 Hz. Spike sorting was done using kilosort, and spikes were manually curated using Phy to remove artifacts picked by the algorithm. Spike times were verified with cross correlograms to combine units or eliminate duplicates. Units “**quality**” was determined using kilosort estimated contamination ratio (“good” units have estimated contamination ratio < 0.2)

Behavioral data: Physiology data was collected at a rate of 30 kHz using the Spike GLX software. Time stamps of each trial start were recorded by the neuropixels Spike GLX software.

Full information on data preprocessing can be found in Drummond et al. (2024) biorXiv. doi: 10.1101/2024.10.24.620009